Momentum analysis: put it together

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a model to compare a deflection option and its limits.

SUCCESS CRITERIA - I can...

1. Compute the required change before evaluating an option.
2. Keep impulse, velocity change and displacement distinct.
3. Communicate assumptions with the numerical result.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Total momentum initially is (15, 0) kg·m/s. One fragment leaves with (3, 5) kg·m/s. Find the other momentum vector and magnitude.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: shift = Δv × time; J = MΔv.
Fictional straight-line planning model. M is target mass; J is required impulse. This is not a real encounter forecast.

Worked reasoning: A 1 kg cart at +6 m/s sticks to a 2 kg cart at rest. Final velocity is +2 m/s. The moving cart changes momentum by -4 and the other by +4 kg·m/s.

Your turn: Draw both momentum changes and show how their sum matches the system change.

Before **After**

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

03 Read and interpret the model

Organize this chain with units: impulse → velocity change → sideways shift. Which step uses target mass?

Impulse J → Change Δv → Offset Δx

Marzano check - circle: ① ② ③

04 Apply the physics

Target mass 2×10^{9} kg; desired shift 20 km over 4×10^{6} s. Find required Δv and impulse. Compare an option delivering 8×10^{6} N·s. Organize the known quantities on the before-and-after model.

Before **After**

Label objects and signed velocities.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

05 Explain your reasoning

If the target mass estimate is too low, how would a fixed delivered impulse's velocity change compare with the prediction?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 4 kg cart moving right at 6 m/s stops in 3 s. Take right as positive. Find signed impulse and average force. Use before and after snapshots to organize the momentum account.

Before	After

Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

07 Exit check

Keep impulse fixed but double both target mass and lead time. What happens to modeled shift? Justify with an equation. Organize the known quantities on the before-and-after model.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 2 kg cart at +4 m/s and a 1 kg cart at -2 m/s stick while an external impulse of -3 N·s acts. Find final velocity and explain why setting initial momentum equal to final would fail. Organize the known quantities on the before-and-after model.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3